Total No. of Questions : 7 [Total No. of Printed Pages : 5

Roll No. ....

REF NO. 067789

B. Sc. (Hons) Semester III Examination 2023 – 24

MATHEMATICS

Paper No. MTB-302 : Differential Equations

Time : 3 hours **Full Marks : 70**

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

Answer five questions including Question 1 which is compulsory.

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1. **Answer all parts of this question :**

(a) Define the order and degree of a differential equation. 2

(b) Obtain the relation in a, b and c (all constants), when $y = xe^{-(b/2a)x}$ is a solution of the differential equation

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

2

(c) Find an integrating factor for the differential equation

$$y\,(axy + e^x)\,dx - e^x\,dy = 0$$

2

(d) Solve the equation

$$\frac{dy}{dx} + ay = e^{mx}$$

where a and m are constants. 2

(e) Obtain the differential equation of the family of plane curves, represented by $cy = 2 + c^2x$. 2

(f) Let $y(x)$ be a solution of

$$\frac{dy}{dx} + 3y = 4, \quad y(0) = 3$$

then find $\lim_{x\to\infty} y(x) = ?$ 2

(g) Find the value of

$$\frac{1}{(D^2 + a^2)^2}\sin(ax)$$

where $D \equiv \frac{d}{dx}$. 2

(h) Find the condition for which $y = x^m$ will be a solution of

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

2

(i) Write a second-order differential equation, which has $x = 1$ as regular singular point and $x = 0$ as an irregular singular point. 2

(j) Express $x^3 + 3x^2$ in terms of Legendre's polynomials. 2

(k) Prove that—

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

2

2. (a) Solve the initial value problem :

$$\frac{dy}{dx} + y \cot x = \cos^3 x \, (1 + y^2 \sin^2 x), \quad y\left(\frac{\pi}{2}\right) = 1$$

6

(b) Solve the differential equation

$$xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$$

where $p = \dfrac{dy}{dx}$. 6

3. (a) Solve :

$$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x$$

6

(b) Solve :

$$\frac{d^2y}{dx^2} + a^2 y = \sec ax$$

6

4. (a) If

$$f(D) = \sum_{r=0}^{n} a_r D^r$$

where a_r $(i = 0, 1, 2, \ldots, n)$ are constants and $D \equiv \dfrac{d}{dx}$, then establish the formula

$$\frac{1}{f(D)}[\cos ax] = \frac{f_1(-a^2)\cos ax + a f_2(-a^2)\sin ax}{f_1^2(-a^2) + a^2 f_2^2(-a^2)}$$

where $f(D) = f_1(D^2) + D f_2(D^2)$. 6

(b) Using the method of variation of parameters, solve the differential equation :

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = x^2 \log x, \quad x > 0$$

6

5. Solve the equation

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} + xy = 0$$

in series form. 12

6. (a) Prove that—

$$P_n(x) = \frac{1}{2^n(n!)}\frac{d^n}{dx^n}(x^2-1)^n$$

where $P_n(x)$ is Legendre polynomials. 7

(b) Prove that—

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$$

where $J_n(x)$ is Bessel's function. 5

7. (a) Show that $P_n(x)$ is the coefficient of z^n is the expansion of $(1-2xz+z^2)^{-1/2}$ in ascending powers of z, where $|x| \leq 1$, $|z| < 1$. 7

(b) Without using any recurrence relation, prove that—

$$J_{3/2}(x) = \sqrt{\frac{2}{\pi x}}\left(\frac{\sin x}{x} - \cos x\right)$$

5

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